Källén–Lehmann and Nambu–Goldstone

Hyeongmuk Lim

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Abstract

The Källén–Lehmann and Nambu–Goldstone theorems are two non-perturbative theorems that apply to all quantum field theories formalized in terms of Wightman axioms. We review the shared ideas behind the two theorems and discuss their proofs.

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1 Introduction

The Källén–Lehmann and Nambu–Goldstone theorems are two structural results that apply to all relativistic quantum field theories (QFTs) in the sense of Wightman axioms. Although they are separate theorems that are often taught in different contexts, the ideas behind the theorems are much in common. The skeleton of the proof of the theorems consists of the following ingredients.

(1) Positivity. The principles of quantum mechanics insist that the Hilbert space is equipped with a positive-definite inner product. As a consequence, the energy-momentum content of any state created from the vacuum is described by a *non-negative* spectral measure.

(2) Spectrum condition. Stability of the vacuum requires that the joint spectrum of the energy-momentum operators lies entirely in the closed forward lightcone: every state has non-negative energy in every Lorentz frame. Hence the spectral measures of (1) are supported in the forward lightcone.

(3) Symmetry. Under these assumptions, symmetries strongly constrain the dynamics of QFT. In the Källén–Lehmann case the symmetry in play is Lorentz symmetry, while for the Nambu–Goldstone theorem an internal symmetry, generated by a conserved current, enters in addition to Lorentz symmetry.

(4) Locality. Causality in the strict sense — fields at spacelike separation commute — plays no role in the Källén–Lehmann representation, but it is essential for the Nambu–Goldstone theorem: it makes the order parameter well defined although the broken charge does not exist as an operator, and it fixes the phase of the Goldstone contribution.

One line of argument. Lorentz invariance decomposes a spectral measure on the forward lightcone into mass shells $p^2 = -s$ (this is the Källén–Lehmann representation, §3). Current conservation multiplies the contribution of the shell s by s; hence it kills every massive shell and leaves only the lightcone $s = 0$. The coefficient of the surviving lightcone term is precisely the order parameter (§4).

Conventions. The spacetime metric is mostly positive $(-,+,+,+)$, $a \cdot b = -a^0 b^0 + \mathbf{a} \cdot \mathbf{b}$, and the Fourier transform is

$$\tilde{f}(p) = \int \mathrm{d}^4x\, e^{-ip\cdot x} f(x), \qquad f(x) = \int \frac{\mathrm{d}^4p}{(2\pi)^4}\, e^{ip\cdot x} \tilde{f}(p). \tag{1}$$

Thus a positive-energy plane wave is $e^{ip\cdot x} = e^{-ip^0 t + i\mathbf{p}\cdot\mathbf{x}}$ with $p^0 > 0$, and a mass shell is $p^2 = -s$ with $s \geq 0$.

2 Wightman axioms

Definition 2.1 (Bosonic QFT)**.** *A* bosonic quantum field theory *is characterized by a Hilbert space $\mathcal{H}$ with a positive-definite inner product, a dense subspace $\mathcal{D} \subset \mathcal{H}$, and the* field operators *$\varphi_a(f)$ for each Schwartz function $f \in \mathcal{S}(\mathbb{R}^4)$ satisfying the following axioms.*

(W1) *$\varphi_a(f)\mathcal{D} \subset \mathcal{D}$, and for each $\psi, \chi \in \mathcal{D}$ the map $f \mapsto \langle \psi, \varphi_a(f)\chi\rangle$ is continuous. We call the field φ_a* Hermitian *if $\varphi_a(\bar{f}) \subset \varphi_a(f)^\dagger$.*

(W2) *The proper orthochronous Poincaré group $\mathcal{P}_+^\uparrow$ has a strongly continuous unitary representation $(a,\Lambda) \mapsto U(a,\Lambda)$ on $\mathcal{H}$, such that for scalar field A and vector field j^μ,*

$$U(a,\Lambda)\, A(f)\, U(a,\Lambda)^{-1} = A\big(f \circ (a,\Lambda)^{-1}\big), \tag{2a}$$

$$U(a,\Lambda)\, j^\mu(f)\, U(a,\Lambda)^{-1} = (\Lambda^{-1})^\mu{}_\nu\, j^\nu\big(f \circ (a,\Lambda)^{-1}\big), \tag{2b}$$

where $\big(f\circ(a,\Lambda)^{-1}\big)(x) = f\big(\Lambda^{-1}(x-a)\big)$. In terms of the point fields of Remark 2.3 below, these read

$$U(a,\Lambda)A(x)U(a,\Lambda)^{-1} = A(\Lambda x + a), \qquad U(a,\Lambda)j^\mu(x)U(a,\Lambda)^{-1} = (\Lambda^{-1})^\mu{}_\nu\, j^\nu(\Lambda x + a).$$

Furthermore, there is a unique one-dimensional invariant subspace, spanned by a unit vector Ω called the vacuum *($\langle\Omega|\Omega\rangle = 1$).*

(W3) The translations $a \mapsto U(a,1)$ form a strongly continuous unitary representation of $\mathbb{R}^4$. By Stone's theorem it has commuting self-adjoint generators P^μ such that $U(a,1) = \exp[-iP\cdot a]$. The joint spectrum $\operatorname{spec} P$ of the generators P^μ is included in the closed forward lightcone $\overline{V}_+ = \{\, p \,|\, p^2 \le 0,\ p^0 \ge 0 \,\}$.

(W4) $[\varphi_a(f), \varphi_b(g)] = 0$ whenever the supports of f, g are spacelike-separated:

$$x \in \operatorname{supp} f,\ y \in \operatorname{supp} g \quad \Longrightarrow \quad (x-y)^2 > 0. \tag{3}$$

There is a fifth axiom (W5) that Ω is cyclic, but we do not use it in this note.

We denote the joint spectral measure of $P = (P^0, \mathbf{P})$ by $E(\cdot)$. Then (W3) states that $E(\mathbb{R}^4 \backslash \overline{V}_+) = 0$. The existence of unique vacuum in (W2) is equivalent to $E(\{0\}) = |\Omega\rangle\langle\Omega|$.

Remark 2.2 (Fermions)**.** *In the presence of fermionic fields one introduces the grading $\Gamma = (-1)^F$ ($\Gamma\Omega = \Omega$) and replaces (W4) by graded locality: two fermionic fields* anticommute *at spacelike separation, while every other pair (boson–boson, boson–fermion) commutes. In addition $\mathcal{P}_+^\uparrow$ is replaced by its double cover. All fields used in this note — scalar fields and conserved currents — are bosonic, so (W4) as stated suffices. Note also that an order parameter must be bosonic: Γ-invariance of the vacuum forces $\langle\Omega|\psi|\Omega\rangle = 0$ for a fermionic ψ, so fermionic symmetry breaking is detected through bosonic composites such as $\bar\psi\psi$.*

Physical meaning of the axioms. (W1) says that a field at a sharp point is too singular to be an operator: already $\langle\Omega|\varphi(x)\varphi(x)|\Omega\rangle$ diverges. Only averages $\varphi(f) = \int \mathrm{d}^4x\, f(x)\varphi(x)$ over smooth, rapidly decaying f are operators, and correlation functions are *tempered distributions*, i.e. continuous linear functionals on $\mathcal{S}$. We write $\mathcal{S}^\vee$ for the space of tempered distributions. Every element of $\mathcal{S}^\vee$ has a Fourier transform in $\mathcal{S}^\vee$, which is why the momentum-space arguments below make sense. (W2) is relativistic covariance, together with the statement that the vacuum is the unique Poincaré-invariant state. (W3) is stability: in every Lorentz frame the energy is bounded below by the vacuum energy, and there are no tachyonic states. (W4) is Einstein causality: measurements in spacelike-separated regions do not disturb each other.

Remark 2.3 (Point fields)**.** *Formally $\varphi_a(f) = \int \mathrm{d}^4x\, f(x)\,\varphi_a(x)$, so that $\varphi_a(x)$ "$= \varphi_a(\delta_x)$" is obtained by inserting the delta function δ_x, which is not a Schwartz function. In this note we compute with the point fields $\varphi_a(x)$ as physicists do. Each identity derived this way is an identity between tempered distributions, and becomes a rigorous statement after smearing all point fields with test functions; we comment on this where it matters.*

Remark 2.4 (Sign of the momentum operator)**.** *With $U(a,1) = e^{-iP\cdot a}$ we have $A(x) = e^{-iP\cdot x}A(0)e^{iP\cdot x}$, i.e. $\partial_\mu A(x) = -i[P_\mu, A(x)]$. In the mostly-plus metric $P_\mu = (-H, \mathbf{P})$, hence*

$$\partial_t A = i[H, A], \qquad \nabla A = -i[\mathbf{P}, A]. \tag{4}$$

The first is the Heisenberg equation. In the second, $\mathbf{P}$ *acts on the operator as* $+i\nabla$*; this is consistent with* $\mathbf{p} = -i\nabla$ *on wavefunctions, because* $\psi(\mathbf{x}) = \langle \mathbf{x}|\psi\rangle$ *involves the complex conjugate of the state* $A(\mathbf{x})|\Omega\rangle \sim |\mathbf{x}\rangle$*.*

Before proceeding, we discuss some direct results of the Wightman axioms which we need in later sections. Recall that the functions of the form $\langle\Omega|A(x)B(y)C(z)\cdots|\Omega\rangle$ are called *correlation functions* of the given QFT. The notation $\langle\mathcal{O}\rangle$ is a shorthand for $\langle\Omega|\mathcal{O}|\Omega\rangle$.

Observation 2.5. *For a pair of fields* A, B *the correlation function* $\langle\Omega|A(x)B(y)|\Omega\rangle$ *depends only on* $x - y$*.*

$$\langle\Omega|A(x)B(y)|\Omega\rangle = w_{AB}(x-y), \qquad w_{AB} \in \mathcal{S}^{\vee}(\mathbb{R}^4). \tag{5}$$

Indeed, translation invariance of Ω *gives*

$$\langle\Omega|A(x+a)B(y+a)|\Omega\rangle = \langle\Omega|U(a,1)A(x)B(y)U(a,1)^{-1}|\Omega\rangle = \langle\Omega|A(x)B(y)|\Omega\rangle.$$

Now consider the correlation function $\langle\Omega|A(x)A(0)|\Omega\rangle$, with A Hermitian. Using a joint eigenbasis $|\lambda\rangle$ for P and $A(x) = U(x,1)\,A(0)\,U(x,1)^{-1} = e^{-iP\cdot x}A(0)e^{iP\cdot x}$,

$$\langle\Omega|A(x)A(0)|\Omega\rangle = \sum_{\lambda}\langle\Omega|A(0)e^{iP\cdot x}|\lambda\rangle\langle\lambda|A(0)|\Omega\rangle = \int e^{i\lambda\cdot x}\mathrm{d}\mu_{A\Omega}(\lambda) \tag{6}$$

In Eq. (6), $\mathrm{d}\mu_{A\Omega}$ is the spectral weight of the state $A(0)|\Omega\rangle$ in the eigenspace E_λ of P, i.e. $\mathrm{d}\mu_{A\Omega}(\lambda) = \langle\Omega|A(0)\,\mathrm{d}E(\lambda)\,A(0)|\Omega\rangle$; the sum over λ includes all degeneracy labels. By definition, $\mathrm{d}\mu_{A\Omega} \geq 0$ and $\mathrm{d}\mu_{A\Omega}(\mathbb{R}^4\backslash\overline{V}_+) = 0$. It is not a probability measure: its total mass $\|A(0)\Omega\|^2 = \langle\Omega|A(0)A(0)|\Omega\rangle$ is typically infinite (an ultraviolet divergence), which is exactly the reason why the rigorous treatment smears A with test functions. Since $\langle\Omega|A(x)A(0)|\Omega\rangle = w_{AA}(x) = \int \frac{\mathrm{d}^4\lambda}{(2\pi)^4}\, e^{i\lambda\cdot x}\,\tilde{w}_{AA}(\lambda)$, we have

$$\mathrm{d}\mu_{A\Omega}(\lambda) = (2\pi)^{-4}\,\tilde{w}_{AA}(\lambda)\,\mathrm{d}^4\lambda. \tag{7}$$

It follows that $\rho_A := \tilde{w}_{AA}$ is a non-negative measure supported on $\overline{V}_+$. More generally, for two Hermitian fields A, B the same insertion of $\sum_\lambda |\lambda\rangle\langle\lambda|$ gives

$$\langle\Omega|A(x)B(0)|\Omega\rangle = \int e^{i\lambda\cdot x}\,\langle\Omega|A(0)\,\mathrm{d}E(\lambda)\,B(0)|\Omega\rangle, \tag{8}$$

$$(2\pi)^{-4}\,\tilde{w}_{AB}(\lambda)\,\mathrm{d}^4\lambda = \langle\Omega|A(0)\,\mathrm{d}E(\lambda)\,B(0)|\Omega\rangle. \tag{9}$$

Above discussions can be put on a mathematically sound basis:

Lemma 2.6. *For two fields* A, B *satisfying (W1)–(W3),*

(1) $\operatorname{supp}\tilde{w}_{AB} \subset \overline{V}_+$*.*

(2) If A *is Hermitian,* $\rho_A = \tilde{w}_{AA}$ *is a non-negative tempered measure.*

(3) If A and B are Hermitian, $\tilde{w}_{AB}$ is a complex measure, and for every Borel set $X \subset \mathbb{R}^4$

$$|\tilde{w}_{AB}(X)|^2 \leq \rho_A(X)\,\rho_B(X). \tag{10}$$

In particular $\tilde{w}_{AB}$ vanishes on every set on which ρ_A (or ρ_B) vanishes.

Physically, (1) is the spectrum condition, (2) is the positivity of the inner product, and (3) is the Cauchy–Schwarz inequality $|\langle u, E(X)v\rangle|^2 \leq \langle u, E(X)u\rangle\langle v, E(X)v\rangle$ applied to $u = A(0)\Omega$ and $v = B(0)\Omega$ in Eq. (9): the field B cannot overlap with A in a momentum region where A creates no states at all. The rigorous proof replaces $A(0)\Omega$, $B(0)\Omega$ by $A(g)\Omega$, $B(h)\Omega$ and uses the uniqueness of the Fourier transform [9, 11]; we omit it.

Imposing Lorentz invariance further constrains the form of $\tilde{w}_{AA}(\lambda)$, since Lorentz boosts and rotations act nontrivially on the momentum space, while acting trivially on the correlation function: It follows from $(W2)$ that

$$\langle\Omega|A(x)A(0)|\Omega\rangle = \langle\Omega|A(\Lambda x)A(\Lambda 0)|\Omega\rangle = \langle\Omega|A(\Lambda x)A(0)|\Omega\rangle. \tag{11}$$

(Insert $A(\Lambda x) = U(\Lambda)A(x)U(\Lambda)^{-1}$ and $A(0) = U(\Lambda)A(0)U(\Lambda)^{-1}$, and use $U(\Lambda)^{-1}\Omega = \Omega$. This uses both that A is a scalar and that the vacuum is Lorentz invariant.) As a result, $\mathrm{d}\mu_{A\Omega}(\lambda) = \mathrm{d}\mu_{A\Omega}(\Lambda\lambda)$ for all $\Lambda \in \mathcal{L}_+^\uparrow$. What can we say about the form of such a measure?

The core idea is that the orbits of $\mathcal{L}_+^\uparrow$ in the forward lightcone $\overline{V}_+$, where $\mathrm{d}\mu_{A\Omega}$ is supported, are

- the hyperboloids (mass shells) $M_s := \left\{\, p \in \overline{V}_+ \,\middle|\, p^2 = -s \,\right\}$ with $s > 0$;
- the punctured lightcone $M_0 := \left\{\, p \in \overline{V}_+ \,\middle|\, p^2 = 0,\ p \neq 0 \,\right\}$;
- the origin $\{0\}$.

Note that M_0 and $\{0\}$ are *different* orbits: a massless particle cannot be boosted to zero momentum. The probability measure smeared uniformly through M_s is

$$\int \frac{\mathrm{d}p^0}{(2\pi)^4}\, 2\pi\,\delta(p^2+s)\,\theta(p^0)\,\mathrm{d}^3p = \frac{\mathrm{d}^3p}{(2\pi)^3 2\omega_{\mathbf{p}}^{(s)}}, \qquad \omega_{\mathbf{p}}^{(s)} = \sqrt{|\mathbf{p}|^2 + s}. \tag{12}$$

Both d^4p and $\delta(p^2+s)\theta(p^0)$ are Lorentz invariant, so this measure is invariant; it is $\mathrm{d}^3p/2\omega$ rather than d^3p that is invariant. The p^0-integral in Eq. (12) is well defined whenever $p^0 = \omega_{\mathbf{p}}^{(s)} > 0$, which holds on every M_s including the punctured lightcone M_0, where it gives $\mathrm{d}^3p/2|\mathbf{p}|$. This measure has no point mass at $\mathbf{p} = 0$, since $\int_{|\mathbf{p}|<\epsilon} \mathrm{d}^3p/2|\mathbf{p}| \to 0$. The origin is a separate orbit, which carries its own invariant measure $c\,\delta^4(p)\,\mathrm{d}^4p$. As a result, any Lorentz-invariant measure ρ can be expressed as

$$\rho(\mathrm{d}^4p) = c\,\delta^4(p)\,\mathrm{d}^4p + 2\pi\int_{[0,\infty)} \mathrm{d}\sigma(s)\,\delta(p^2+s)\,\theta(p^0)\,\mathrm{d}^4p. \tag{13}$$

The two massless terms must not be confused: c is the weight of the single point $p = 0$ (we will see that it is the vacuum contribution), while $\sigma(\{0\})$ is the weight of the punctured lightcone M_0 (massless particles with $p \neq 0$). This observation can also be formulated in mathematical rigor:

Lemma 2.7. *Let ρ be a Lorentz-invariant measure on $\overline{V}_+$, i.e., $\rho(\mathcal{L}_+^\uparrow X) = \rho(X)$. Assume that ρ is non-negative and tempered. Then there exist a unique $c \geq 0$ and a unique non-negative measure σ on $[0,\infty)$ such that Eq. (13) holds. Here $c = \rho(\{0\})$, and $\int \mathrm{d}\sigma(s)\,(1+s)^{-N} < \infty$ for some N.*

Sketch of proof. Restrict ρ to each $\mathcal{L}_+^\uparrow$-invariant piece. The origin contributes $c = \rho(\{0\})$. On $\overline{V}_+\backslash\{0\}$, write $p = \sqrt{s}\,u$ on the hyperboloids, and use the parametrization $(s, \mathbf{p})$ on M_0. For each Borel set $S \subset [0,\infty)$, the measure $X \mapsto \rho\big(X \cap \bigcup_{s\in S} M_s\big)$ is an invariant measure on a single orbit type. Each orbit M_s is a homogeneous space $\mathcal{L}_+^\uparrow/K$ (with $K = SO(3)$ for $s > 0$ and $K = ISO(2)$ for $s = 0$), and its invariant measure is unique up to a constant: it is Eq. (12) [9, §IX.8]. The constant, as a function of S, defines $\sigma(S)$. Temperedness of ρ translates into the growth bound on σ. □

3 The Källén–Lehmann representation

For $s \geq 0$ let $D(x;s)$ denote the Wightman function of a free scalar field of mass $\sqrt{s}$,

$$D(x;s) = \int \frac{\mathrm{d}^3p}{(2\pi)^3 2\omega_{\mathbf{p}}^{(s)}}\, e^{-i\omega_{\mathbf{p}}^{(s)}t + i\mathbf{p}\cdot\mathbf{x}} = \int \frac{\mathrm{d}^4p}{(2\pi)^4}\, e^{ip\cdot x}\, 2\pi\,\delta(p^2+s)\,\theta(p^0), \tag{14}$$

and write $D_0(x) := D(x;0)$ and $\Delta(x;s) := D(x;s) - D(-x;s)$ (the vacuum expectation value of the free-field commutator). We record two facts used repeatedly.

- $\Delta(x;s) = 0$ for spacelike x. (It is Lorentz invariant and odd under $x \to -x$, and for spacelike x some $\Lambda \in \mathcal{L}_+^\uparrow$ maps x to $-x$.)
- At equal times, $\partial_t \Delta(x;s)\big|_{t=0} = -i\,\delta^3(\mathbf{x})$, independently of s. Indeed

$$\partial_t \Delta(x;s)\big|_{t=0} = \int \frac{\mathrm{d}^3p}{(2\pi)^3 2\omega_{\mathbf{p}}^{(s)}}\,(-i\omega_{\mathbf{p}}^{(s)})\big(e^{i\mathbf{p}\cdot\mathbf{x}} + e^{-i\mathbf{p}\cdot\mathbf{x}}\big) = -i\,\delta^3(\mathbf{x}).$$

For spacelike x the massless function is $D_0(x) = D_0(-x) = 1/(4\pi^2x^2)$, which is easily checked at $t = 0$.

Theorem 3.1 (Källén–Lehmann)**.** *Let φ be a Hermitian scalar field satisfying (W1)–(W3). Then there exists a unique non-negative measure σ on $[0,\infty)$ such that*

$$\langle\Omega|\varphi(x)\varphi(0)|\Omega\rangle = \langle\varphi\rangle^2 + \int \mathrm{d}\sigma(s)\,\frac{\mathrm{d}^3p}{(2\pi)^3 2\omega_{\mathbf{p}}^{(s)}}\, e^{-i\omega_{\mathbf{p}}^{(s)}t + i\mathbf{p}\cdot\mathbf{x}} = \langle\varphi\rangle^2 + \int_{[0,\infty)} \mathrm{d}\sigma(s)\, D(x;s). \tag{15}$$

Proof. Insert $1 = E(\{0\}) + E(\overline{V}_+\backslash\{0\})$ into $\langle\Omega|\varphi(x)\varphi(0)|\Omega\rangle = \langle\Omega|\varphi(0)\, e^{iP\cdot x}\,\varphi(0)|\Omega\rangle$.

The point $p = 0$. Since $E(\{0\}) = |\Omega\rangle\langle\Omega|$ and $e^{iP\cdot x}E(\{0\}) = E(\{0\})$, the first piece is the x-independent constant

$$\langle\Omega|\varphi(0)|\Omega\rangle\langle\Omega|\varphi(0)|\Omega\rangle = \langle\varphi\rangle^2, \tag{16}$$

real because φ is Hermitian. In terms of Eq. (13) for $\rho_\varphi = \tilde{w}_{\varphi\varphi}$, this says $c = (2\pi)^4\langle\varphi\rangle^2$: the weight of the atom at $p = 0$ is *not* a value of a density, but the squared norm $\|E(\{0\})\varphi(0)\Omega\|^2$ of the vacuum component of $\varphi(0)\Omega$.

The rest. The remaining piece

$$w^{\mathrm{c}}(x) = \int_{\overline{V}_+\setminus\{0\}} e^{i\lambda\cdot x}\,\mathrm{d}\mu_{\varphi\Omega}(\lambda) \tag{17}$$

is the Fourier transform of $(2\pi)^{-4}\rho_\varphi$ restricted to $\overline{V}_+\setminus\{0\}$, a Lorentz-invariant non-negative measure with no weight at the origin. By Lemma 2.7 (with $c = 0$),

$$\rho_\varphi\big|_{\overline{V}_+\setminus\{0\}}(\lambda)\,\mathrm{d}^4\lambda = 2\pi\int_{[0,\infty)} \mathrm{d}\sigma(s)\,\delta(\lambda^2 + s)\,\theta(\lambda^0)\,\mathrm{d}^4\lambda, \tag{18}$$

and by Eq. (14) the inverse Fourier transform is $w^{\mathrm{c}}(x) = \int \mathrm{d}\sigma(s)\,D(x;s)$. Uniqueness of σ follows from the uniqueness in Lemma 2.7. □

Remark 3.2 (Volume averaging)**.** *The constant can also be extracted by averaging over spacetime:*

$$\begin{aligned}\lim_{V\to\infty}\frac{1}{V}\int_V \mathrm{d}^4x\,\langle\Omega|\varphi(x)\varphi(0)|\Omega\rangle &= \Big\langle\Omega\Big|\varphi(0)\Big[\lim_{V\to\infty}\frac{1}{V}\int_V \mathrm{d}^4x\,e^{iP\cdot x}\Big]\varphi(0)\Big|\Omega\Big\rangle \\ &= \langle\Omega|\varphi(0)E(\{0\})\varphi(0)|\Omega\rangle = \langle\varphi\rangle^2.\end{aligned} \tag{19}$$

The bracket converges to the projection $E(\{0\})$ onto $P = 0$ (von Neumann's mean ergodic theorem): the continuous part of the spectrum oscillates and averages out, and only the atom survives. Equivalently, $D(x;s) \to 0$ as $x \to \infty$ in spacelike directions, so $\langle\Omega|\varphi(x)\varphi(0)|\Omega\rangle \to \langle\varphi\rangle^2$, which is the cluster property. Note that the average divides $\int_V \mathrm{d}^4x$ by V; the value $\rho_\varphi(0)$ of the Fourier transform at a single point is not a meaningful quantity for a measure with an atom.

Remark 3.3 (c versus $\sigma(\{0\})$)**.** *Both the constant $\langle\varphi\rangle^2$ and the term $\sigma(\{0\})D_0(x)$ are "massless", but they are different: the first comes from the vacuum ($p = 0$), the second from massless particles ($p^2 = 0$, $p \neq 0$). The Nambu–Goldstone theorem is a statement about the second.*

Corollary 3.4 (Sum rule and $Z \leq 1$)**.**

(1) $\langle\Omega|[\varphi(x),\varphi(0)]|\Omega\rangle = \int \mathrm{d}\sigma(s)\,\Delta(x;s)$*; the constant term drops out.*

(2) If φ obeys the canonical equal-time commutation relation $[\varphi(0,\mathbf{x}),\partial_t\varphi(0,\mathbf{y})] = i\,\delta^3(\mathbf{x}-\mathbf{y})$, then $\int_{[0,\infty)}\mathrm{d}\sigma(s) = 1$.

(3) Consequently, if σ contains a one-particle atom $Z\,\delta(s-m^2)$, then $0 \leq Z \leq 1$, and $Z = 1$ forces $\sigma = \delta(s-m^2)$, i.e. the two-point function of a free field.

Proof. (1) follows from Eq. (15) and $\langle\Omega|\varphi(0)\varphi(x)|\Omega\rangle = \overline{\langle\Omega|\varphi(x)\varphi(0)|\Omega\rangle}$. For (2), apply $\partial_t|_{t=0}$ to (1) and use $\partial_t\Delta|_{t=0} = -i\delta^3(\mathbf{x})$: the right-hand side becomes $-i\,\delta^3(\mathbf{x})\int\mathrm{d}\sigma$, while the left-hand side is $\langle\Omega|[\partial_t\varphi(0,\mathbf{x}),\varphi(0)]|\Omega\rangle = -i\,\delta^3(\mathbf{x})$. (3) follows from $\sigma \geq 0$. □

4 The Nambu–Goldstone theorem

Nambu observed that the Bogoliubov quasiparticles of a superconductor come with a collective mode tied to the broken phase symmetry [8], and Goldstone conjectured that the spontaneous breaking of a continuous symmetry always produces massless particles [3]. After the first general argument of Goldstone, Salam and Weinberg [4], the statement was turned into a theorem of axiomatic QFT by Streater [10], Kastler, Robinson and Swieca [6], and Ezawa and Swieca [2]. In this section we give the relativistic version of the proof in the spirit of the Källén–Lehmann representation, computing with point fields throughout (Remark 2.3).

4.1 Conserved currents and order parameters

Throughout this section, j^μ is a Hermitian vector field and A is a Hermitian scalar field, both satisfying (W1)–(W4), and the current is conserved:

$$\partial_\mu j^\mu(x) = 0. \tag{20}$$

Formally the current generates the symmetry through the charge $Q = \int \mathrm{d}^3x\, j^0(t,\mathbf{x})$, and the infinitesimal transformation of A is $\delta A = i[Q, A]$ (so that $e^{i\theta Q} A e^{-i\theta Q} = A + \theta\,\delta A + O(\theta^2)$). The symmetry is *unbroken* if $Q\Omega = 0$, and then $\langle\Omega|[Q, A(0)]|\Omega\rangle = 0$ for every A. Spontaneous breaking means that some field A has $\langle\Omega|\delta A|\Omega\rangle \neq 0$: the vacuum is not invariant.

Caution 4.1. *If Q were a self-adjoint operator with Ω in its domain, then $Q\Omega$ would be translation invariant ($[P^\mu, Q] = 0$), hence $Q\Omega = c\,\Omega$ with $c \in \mathbb{R}$ by uniqueness of the vacuum, and $\langle\Omega|[Q, A(0)]|\Omega\rangle = c\langle A\rangle - c\langle A\rangle = 0$. So when the symmetry is spontaneously broken, Q does* not *exist as an operator on $\mathcal{H}$: the volume integral of j^0 applied to Ω diverges (formally $\|Q\Omega\| = \infty$). We therefore define the order parameter without Q.*

Definition 4.2 (Order parameter)**.**

$$\eta := \int \mathrm{d}^3x\, \langle\Omega|[j^0(0,\mathbf{x}), A(0)]|\Omega\rangle. \tag{21}$$

The symmetry generated by j^μ is spontaneously broken *if $\eta \neq 0$ for some A.*

Two properties make this definition sensible, and both come from locality.

- *The integral is local.* By (W4), $[j^0(0,\mathbf{x}), A(0)] = 0$ for $\mathbf{x} \neq 0$, since the two points are spacelike separated. The integrand is a distribution supported at $\mathbf{x} = 0$, so Eq. (21) only involves an arbitrarily small neighbourhood of the origin. The infrared divergence of Q never enters.

- *It does not depend on the time slice.* Let $\eta(t) = \int \mathrm{d}^3x\, \langle\Omega|[j^0(t,\mathbf{x}), A(0)]|\Omega\rangle$. By locality the integrand is supported in the ball $|\mathbf{x}| \leq |t|$, so we may integrate over a ball B_R with $R > |t|$. Then, by Eq. (20),

$$\frac{\mathrm{d}\eta}{\mathrm{d}t} = \int_{B_R} \mathrm{d}^3x\, \langle\Omega|[\partial_t j^0, A(0)]|\Omega\rangle = -\oint_{\partial B_R} \mathrm{d}\mathbf{S}\cdot\langle\Omega|[\mathbf{j}(t,\mathbf{x}), A(0)]|\Omega\rangle = 0, \tag{22}$$

because the sphere ∂B_R is spacelike to the origin. Thus η expresses the conservation of the (non-existent) charge.

Finally, η is purely imaginary: since j^0 and A are Hermitian, $\overline{\langle\Omega|[j^0, A]|\Omega\rangle} = \langle\Omega|[A, j^0]|\Omega\rangle = -\langle\Omega|[j^0, A]|\Omega\rangle$. Accordingly $\langle\Omega|\delta A|\Omega\rangle = i\eta$ is real.

4.2 The theorem

Theorem 4.3 (Nambu–Goldstone)**.** *Under the assumptions above,*

$$\langle\Omega|j^\mu(x)A(0)|\Omega\rangle = -i\eta\,\partial^\mu D_0(x), \tag{23a}$$

$$\langle\Omega|[j^\mu(x), A(0)]|\Omega\rangle = -i\eta\,\partial^\mu \Delta(x;0). \tag{23b}$$

In particular, if $\eta \neq 0$, the energy-momentum spectrum contains massless states: there are states $|\lambda\rangle$ with $\lambda^2 = 0$, $\lambda \neq 0$, such that $\langle\Omega|j^\mu(0)|\lambda\rangle \neq 0$ and $\langle\lambda|A(0)|\Omega\rangle \neq 0$. Moreover the spectral measure σ_A of A in the Källén–Lehmann representation has $\sigma_A(\{0\}) > 0$.

The proof follows the line of the Källén–Lehmann representation, with one new input at each step.

Step	Input	Output
1	spectrum condition, positivity	spectral representation of $\langle\Omega\|j^\mu(x)A(0)\|\Omega\rangle$
2	Lorentz covariance	each mass shell contributes $\propto \lambda^\mu$
3	current conservation	only the lightcone $s = 0$ survives
4	locality	the coefficient is purely imaginary
5	equal-time limit	the coefficient equals η

Step 1: spectral representation. Let $W^\mu(x) := \langle\Omega|j^\mu(x)A(0)|\Omega\rangle$. Inserting a complete set of energy-momentum eigenstates as in Eq. (9),

$$W^\mu(x) = \sum_\lambda e^{i\lambda\cdot x}\,\langle\Omega|j^\mu(0)|\lambda\rangle\langle\lambda|A(0)|\Omega\rangle = \int \frac{\mathrm{d}^4\lambda}{(2\pi)^4}\, e^{i\lambda\cdot x}\,\widetilde{W}^\mu(\lambda). \tag{24}$$

By Lemma 2.6, $\widetilde{W}^\mu$ is a complex measure supported in $\overline{V}_+$, and it vanishes wherever ρ_A vanishes.

Step 2: Lorentz covariance. By (W2) and the invariance of Ω, exactly as for the scalar two-point function,

$$W^\mu(\Lambda x) = \Lambda^\mu{}_\nu\, W^\nu(x), \qquad \text{hence} \qquad \widetilde{W}^\mu(\Lambda\lambda) = \Lambda^\mu{}_\nu\, \widetilde{W}^\nu(\lambda). \tag{25}$$

Fix λ and consider the Lorentz transformations that leave λ fixed (the little group). They must leave the vector $\widetilde{W}^\mu(\lambda)$ fixed as well.

- $\lambda \in M_s$, $s > 0$. In the rest frame $\lambda = (\sqrt{s}, \mathbf{0})$ the little group is $SO(3)$, and a rotation-invariant 4-vector has no spatial part. Hence $\widetilde{W}^\mu(\lambda) \propto \lambda^\mu$.

- $\lambda \in M_0$. Take $\lambda = (\omega, 0, 0, \omega)$. The little group $ISO(2)$ consists of rotations about the z-axis and two null rotations. Rotation invariance forces $v = (v^0, 0, 0, v^3)$; a null rotation shifts the transverse components of v by an amount proportional to $\lambda \cdot v$, so invariance forces $\lambda \cdot v = 0$, i.e. $v^0 = v^3$. Hence again $\widetilde{W}^\mu(\lambda) \propto \lambda^\mu$.

- $\lambda = 0$. The little group is all of $\mathcal{L}_+^\uparrow$, which fixes no nonzero vector. Physically, the $\lambda = 0$ term is $\langle\Omega|j^\mu(0)|\Omega\rangle\langle\Omega|A(0)|\Omega\rangle$, and a Lorentz-invariant vacuum has $\langle\Omega|j^\mu|\Omega\rangle = 0$.

The proportionality factor is a Lorentz-invariant function of λ, i.e. a function of $s = -\lambda^2$ only. Hence there is a complex measure τ on $[0, \infty)$ such that

$$\widetilde{W}^\mu(\lambda) = 2\pi\, \lambda^\mu\, \theta(\lambda^0) \int_{[0,\infty)} \mathrm{d}\tau(s)\, \delta(\lambda^2 + s), \tag{26a}$$

$$W^\mu(x) = -i \int_{[0,\infty)} \mathrm{d}\tau(s)\, \partial^\mu D(x; s), \tag{26b}$$

where we used $\partial^\mu e^{i\lambda\cdot x} = i\lambda^\mu e^{i\lambda\cdot x}$ and Eq. (14). By the last statement of Step 1, $\tau(\{0\}) \neq 0$ is possible only if $\rho_A(M_0) > 0$, i.e. only if $\sigma_A(\{0\}) > 0$.

Step 3: current conservation. Differentiating, $\partial_\mu W^\mu(x) = \langle\Omega|\partial_\mu j^\mu(x)\, A(0)|\Omega\rangle = 0$. Since $\partial_\mu \partial^\mu D(x; s) = s\, D(x; s)$ (each plane wave on M_s has $-\lambda^2 = s$),

$$0 = \partial_\mu W^\mu(x) = -i \int \mathrm{d}\tau(s)\, s\, D(x; s), \quad \text{or equivalently} \tag{27a}$$

$$0 = \lambda_\mu \widetilde{W}^\mu(\lambda) = -2\pi\, \theta(\lambda^0) \int \mathrm{d}\tau(s)\, s\, \delta(\lambda^2 + s). \tag{27b}$$

Distinct mass shells are disjoint, so the measure $s\,\mathrm{d}\tau(s)$ vanishes: τ is supported at $s = 0$. Writing $\tau = a\,\delta_0$ with $a \in \mathbb{C}$,

$$W^\mu(x) = -i\, a\, \partial^\mu D_0(x). \tag{28}$$

This is the field-theoretic version of a familiar one-particle argument: if a state of mass m has $\langle\Omega|j^\mu(0)|p\rangle = iF\, p^\mu$, then $\partial_\mu j^\mu = 0$ gives $-F\, p^2 = F\, m^2 = 0$, so either $F = 0$ or $m = 0$.

Step 4: locality fixes the phase. Since j^μ and A are Hermitian, $\langle\Omega|A(0)j^\mu(x)|\Omega\rangle = \overline{W^\mu(x)}$, and $\overline{D_0(x)} = D_0(-x)$. Write $a = a_R + ia_I$. Then

$$\begin{aligned}\langle\Omega|[j^\mu(x), A(0)]|\Omega\rangle &= -i\, a\, \partial^\mu D_0(x) - i\, \bar{a}\, \partial^\mu \big[D_0(-x)\big] \\ &= -i\, a_R\, \partial^\mu \big[D_0(x) + D_0(-x)\big] + a_I\, \partial^\mu \Delta(x; 0).\end{aligned} \tag{29}$$

For spacelike x the left-hand side vanishes by (W4), and so does $\Delta(x; 0)$. But there $D_0(x) + D_0(-x) = 1/(2\pi^2 x^2)$, whose gradient $-x^\mu/\big(\pi^2 (x^2)^2\big)$ is nonzero. Hence $a_R = 0$ and

$$\langle\Omega|[j^\mu(x), A(0)]|\Omega\rangle = a_I\, \partial^\mu \Delta(x; 0). \tag{30}$$

Step 5: the coefficient is the order parameter. Take $\mu = 0$ at $t = 0$. With $\partial^0 = -\partial_t$ and $\partial_t \Delta|_{t=0} = -i\delta^3(\mathbf{x})$,

$$\langle \Omega | [j^0(0,\mathbf{x}), A(0)] | \Omega \rangle = -a_I \, \partial_t \Delta(x;0)\big|_{t=0} = i \, a_I \, \delta^3(\mathbf{x}). \tag{31}$$

Integrating over $\mathbf{x}$ and comparing with Eq. (21) gives $\eta = i a_I = a$. Substituting into Eq. (28) and the previous display gives Eq. (23a) and Eq. (23b). Finally, if $\eta \neq 0$ then $\tau(\{0\}) = \eta \neq 0$, so by Step 2 $\sigma_A(\{0\}) > 0$. Moreover the integrand $\langle \Omega | j^\mu(0) | \lambda \rangle \langle \lambda | A(0) | \Omega \rangle$ of Step 1 is nonzero on the lightcone M_0. □

Corollary 4.4 (Mass gap excludes spontaneous breaking)**.** *If the spectrum of P contains no states with $p^2 = 0$, $p \neq 0$ — for example if $\operatorname{spec}(-P^2) \subset \{0\} \cup [m^2, \infty)$ with $m > 0$ and $p = 0$ only for the vacuum — then $\eta = 0$ for every conserved current j^μ and every scalar field A.*

4.3 Examples and remarks

Example 4.5 (Free massless scalar)**.** *Let ϕ be a free massless scalar with $\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi$, invariant under the shift $\phi \to \phi + \theta$. The Noether current is $j^\mu = -\partial^\mu\phi$, so $j^0 = \partial_t\phi = \pi$ and $[j^0(0,\mathbf{x}), \phi(0)] = -i\delta^3(\mathbf{x})$. Hence $\eta = -i$, i.e. $\langle\Omega|\delta\phi|\Omega\rangle = 1$, even though nothing forces $\langle\Omega|\phi|\Omega\rangle \neq 0$. The order parameter is the expectation value of the* variation *δA, not of A. The theorem predicts $\langle\Omega|j^\mu(x)\phi(0)|\Omega\rangle = -i\eta\,\partial^\mu D_0 = -\partial^\mu D_0(x)$, which is exactly $-\partial^\mu\langle\Omega|\phi(x)\phi(0)|\Omega\rangle$. The Goldstone boson is ϕ itself.*

Example 4.6 (Complex scalar with a Mexican-hat potential)**.** *Let $\Phi = (\phi_1 + i\phi_2)/\sqrt{2}$ with $\mathcal{L} = -\partial_\mu\Phi^*\partial^\mu\Phi - V(|\Phi|)$, and suppose $\langle\Omega|\phi_1|\Omega\rangle = v \neq 0$. The $U(1)$ current is $j^\mu = \phi_2\partial^\mu\phi_1 - \phi_1\partial^\mu\phi_2$, with $j^0 = \phi_1\pi_2 - \phi_2\pi_1$. Taking $A = \phi_2$ gives $[j^0(0,\mathbf{x}), \phi_2(0)] = -i\,\phi_1(0)\,\delta^3(\mathbf{x})$, so $\eta = -iv$. The theorem then says that ϕ_2 creates a massless state from the vacuum, which is the Goldstone boson. Note that the proof never used the potential, perturbation theory, or the existence of Q.*

Remark 4.7 (Why massive shells are forbidden: the charge would oscillate)**.** *Suppose a shell $s > 0$ contributed to the commutator with a coefficient b, i.e. a term $b\,\partial^\mu\Delta(x;s)$ in Eq. (23b). Its contribution to $\eta(t)$ would be*

$$\int \mathrm{d}^3x \; b\,\partial^0\Delta(x;s) = -b\,\partial_t\left[\frac{-i\sin(\sqrt{s}\,t)}{\sqrt{s}}\right] = i\,b\cos(\sqrt{s}\,t), \tag{32}$$

which oscillates in time and contradicts the time independence of η shown above. Only $s = 0$ gives a constant. Hence, with $\partial_\mu j^\mu = 0$, a nonzero order parameter $\eta \neq 0$ can come only from the $s = 0$ term: the field A must create massless particles from the vacuum.

Remark 4.8 (Where each assumption was used)**.**

- Spectrum condition. *It confines $\widetilde{W}^\mu$ to $\overline{V}_+$ (Step 1). This makes the decomposition into mass shells possible, and it gives the explicit form of D_0 used in Step 4.*

- Positivity. *It guarantees that the lightcone term corresponds to genuine states in $\mathcal{H}$ ($\sigma_A(\{0\}) > 0$). In covariant gauges of gauge theories the state space has an indefinite metric, and the Goldstone pole can sit entirely on unphysical states. This is one way to understand the Higgs mechanism.*

- Lorentz covariance. *It forces $\widetilde{W}^\mu \propto \lambda^\mu$ (Step 2). Without it the theorem survives in a weaker form, "no energy gap at long wavelengths" — the remark above only used the conservation law. But the dispersion relation and the number of gapless modes are no longer fixed. For example, a Heisenberg ferromagnet breaks $SO(3) \to SO(2)$ with a single magnon of quadratic dispersion. The general counting rule is due to Watanabe and Murayama [12].*

- Locality. *It makes η well defined and conserved, and it fixes the phase in Step 4. Non-local formulations, such as Coulomb gauge in gauge theories, evade the theorem in the same way as the Higgs mechanism.*

- Dimension. *In two spacetime dimensions the invariant measure on the lightcone, $\mathrm{d}p^1/2|p^1|$, is a measure which is not locally finite at $p = 0$ (equivalently, D_0 is logarithmically infrared divergent). It therefore cannot be part of the positive tempered measure ρ_A, so $\sigma_A(\{0\}) = 0$, and by Step 2 $\eta = 0$: a continuous symmetry cannot be spontaneously broken in a relativistic QFT in $1+1$ dimensions (Coleman [1]). Positivity is essential here, through Lemma 2.6(3).*

Remark 4.9 (Rigour)**.** *Every step above is an identity between tempered distributions, and becomes rigorous after smearing. The equal-time integral Eq. (21) needs more care: one replaces it by $\lim_{R\to\infty}\langle\Omega|[j^0(f_R), A(g)]|\Omega\rangle$ with $f_R(x) = \chi(x^0)h(\mathbf{x}/R)$, where $\int\chi = 1$ and $h(\mathbf{0}) = 1$. The Fourier-space form of Eq. (23b) shows that this limit equals $\eta\int g$. The rigorous statements are in [2, 6], which work in the algebraic framework and do not even assume covariant fields.*

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